

Date: 2026-05-04

Request ID: 100048499

Hotplate		
Element(s)	Al	
SOP#	HF-off	
Cocktail	25 mL HF then 50 mL HCl	
sample(s)	weight (g)	volume (mL)
200127806_1	0.1019	250
200127806_2	0.1058	250

Date: 2026-05-04
Request ID: 100048499

LOI temp: 900 °C

sample	crucible (g)	crucible + sample (g)	after 900 °C	LOI (%)	correction
200127806	27.0137	29.0304	28.898	6.57	0.9343

Al ICP analysis

sample	Dilution	conc. [mg/L]	Al_ppm [dried]	%Al [dried]
200127806_1	50/6	1.703	37264.09	3.73
200127806_2	50/6	1.769	37281.40	3.73

al result: 37272.74 ppm

al result: 3.73 %

al oxide factor: 1.889

al oxide result: 7.04 %

al lot: *Manufacturer: Assurance, 27-171ALT exp: 2026-07-30*

Note(s):

A = crucible

B = crucible + sample

C = crucible + sample after drying

$$\%LOI = ([B-A] - [C-A]) / (B-A) * 100\%$$

$$ppm\ M+ = [conc.][volume][dilution] / [weight]$$

$$\%M+ = [conc.][volume][dilution] / ([weight] * 10,000)$$

$$\%MO = [oxide\ factor] * \%M+$$

$$ppm\ M+ \ [dried] = ppm\ M+ / ([1 - (\%LOI) / 100])$$

$$\%M+ \ [dried] = \%M+ / ([1 - (\%LOI) / 100])$$

$$\%MO \ [dried] = \%MO / ([1 - (\%LOI) / 100])$$

$$\%Cr(VI) = (1.733[mL\ FAS][N\ FAS]) / ([weight])$$

$$\%Cr_2O_3 = 1.462 * [\%Cr(VI)]$$

$$\%Cr(III) = total_Cr - Cr(VI)$$